

NEUROSCI 366

Lecture 20

中央模式发生器与振荡 / Central Pattern Generators and Oscillations

Source-aligned · Static · Searchable · Printable

Source files

- [Lecture20-CentralPatternGenerators.pdf](#)

Production version: 2026-08-22

Reader profile: neuroscience undergraduate education; mathematics scaffolded from high-school algebra/trigonometry to the formal computational-neuroscience level.

This companion preserves source-page order, distinguishes source material from necessary scaffolding and extensions, and places hints/answers after the lesson. It is a static PDF: no forms, JavaScript, hidden-answer widgets, passwords, or DRM.

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1 How to use this companion

1. 先用 5-minute diagnostic 暴露前置缺口；不要立即翻答案。
2. 按原笔记页序阅读 source-page units：先看原页，再读 clean reconstruction 与补全逻辑。
3. 遇到公式按“问题 → 符号/shape/units → assumptions → 一行一步推导 → intuition → biological meaning → sanity check → failure”阅读。
4. 完成 Cumulative Knowledge Check；只在需要时依次打开 Hint 1、2、3。
5. 至少隔一个 page break 后再核对 Answer Key，并记录错误类型：concept、symbol、algebra、assumption、correlation/causation。

Source hierarchy. 原笔记是内容权威；本书中的“【必要前置】/【补全推导】/【跨课连接】/【延伸】”是为了补齐数学或迁移，不替代源材料。无法确认的内容必须进入 Uncertainty Log。

2 Learning objectives and prerequisite dependency map

- 定义并区分：pacemaker vs follower。
- 从第一原则推导：half-center。
- 解释几何/计算直觉：complex eigenvalues。
- 连接生物学解释：Morris–Lecar。
- 诊断 assumptions 与 failure modes：nullclines/limit cycle。
- 把概念迁移到新神经科学问题：release vs escape。

Dependency map

pacemaker vs follower → half-center → complex eigenvalues → Morris–Lecar → nullclines/limit cycle → release vs escape

Core question

rhythm 是 cell-intrinsic 还是 network-generated, linear complex modes、Morris–Lecar fast-slow geometry 与 half-center inhibition 如何分别产生振荡？

3 Five-minute prerequisite diagnostic

1. synapse block 后 rhythm 保留支持什么？（卡住时先看 Source-page unit 1，再看补全推导。）
2. half-center 的核心 connectivity？（卡住时先看 Source-page unit 2，再看补全推导。）
3. complex eigenvalue 的 real part 决定什么？（卡住时先看 Source-page unit 3，再看补全推导。）
4. imaginary part 决定什么？（卡住时先看 Source-page unit 4，再看补全推导。）
5. 为什么笔记线性 network 不产生 sustained rhythm？（卡住时先看 Source-page unit 5，再看补全推导。）

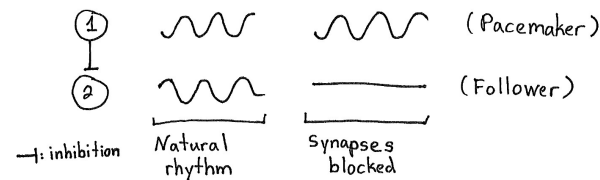
Scoring guide: 4–5 confident answers → proceed; 2–3 → use the prerequisite labels; 0–1 → read the formula/notation sheet once before the source-aligned lesson.

4 Source-aligned lesson / 按原笔记页序

4.1 Lecture20-CentralPatternGenerators.pdf • p. 1

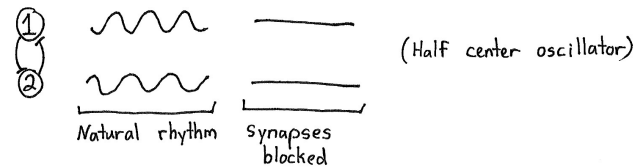
Central Pattern Generators:

As mentioned in motor systems introduction, rhythmic motions of the body often follow rhythmic oscillations of neurons. Some rhythms are generated cell intrinsically.



A well-known example is respiration.

Other rhythms are generated by the network itself. The basic circuit motif is mutual inhibition



A well-known example is muscle alternation during walking.

We want to understand the neural mechanisms behind both rhythm types. We'll see that nonlinearity is essential to get realistic patterns.

Original-note anchor. [Source: Lecture20-CentralPatternGenerators.pdf, p. 1]

Clean reconstruction. Page 1: intrinsic pacemaker/follower, network-generated half-center oscillator, respiration/locomotion.

What the note is saying. 本页的中心对象是 pacemaker vs follower 与 half-center。原笔记将它们放进以下链条：intrinsic pacemaker/follower、network-generated half-center oscillator、respiration/locomotion。

Why it follows. 分析 mutually inhibitory half-center with synaptic threshold; 逐步区分 intrinsic release 和 intrinsic escape。其逻辑后果是：比较 cell-intrinsic pacemaker 与 network-generated rhythm; 解释 synapse block 的诊断逻辑。后面的正式推导会逐项写出 assumptions、shape/units、limiting cases 与 failure conditions。

Figure reading. 逐条追踪箭头方向、兴奋/抑制符号、状态变量与外部输入；结构图说明允许的依赖关系，不自动证明生物实现。

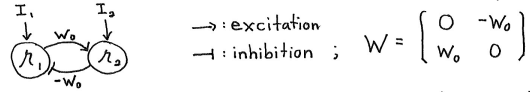
Stop & Predict

synapse block 后 rhythm 保留支持什么？

4.2 Lecture20-CentralPatternGenerators.pdf • p. 2

Oscillations: Central Pattern Generators (Linear)

Let's consider a different two-neuron example.



The first step of our program is to find eigenvalues

$$0 = \det \begin{bmatrix} \lambda & w_0 \\ -w_0 & \lambda \end{bmatrix} = \lambda^2 + w_0^2 \Rightarrow \lambda_{\pm} = \pm i \sqrt{w_0^2} = \pm i |w_0|$$

The eigenvalues are imaginary! Suspending disbelief,

$$\tau \frac{d\alpha_{\pm}}{dt} = -(1 \pm i |w_0|) \alpha_{\pm} + \beta_{\pm}$$

$$\tau_{\text{eff}, \pm} = \frac{\tau}{1 \pm i |w_0|} \Rightarrow e^{-t/\tau_{\text{eff}, \pm}} = e^{-t(1 \pm i |w_0|)/\tau}$$

$$= \underbrace{e^{-t/\tau}}_{\text{exponential decay}} \underbrace{e^{\pm i \frac{|w_0|}{\tau} t}}_{\text{What's this? (imaginary part)}}$$

Fact about complex numbers:

$$e^{i\theta} = \cos(\theta) + i \sin(\theta)$$

$$\Rightarrow e^{\pm i \frac{|w_0|}{\tau} t} = \underbrace{\cos\left(\frac{|w_0|}{\tau} t\right) \pm i \sin\left(\frac{|w_0|}{\tau} t\right)}_{\text{oscillations!}}$$

When the weight matrix has complex eigenvalues, this means that there are oscillatory components to the dynamics, but the math becomes a bit cumbersome, and we won't spend more time on analyzing oscillations analytically in linear systems.

Original-note anchor. [Source: Lecture20-CentralPatternGenerators.pdf, p. 2]

Clean reconstruction. Page 2: linear two-neuron oscillator, complex eigenvalues, Euler formula, damped oscillations.

What the note is saying. 本页的中心对象是 complex eigenvalues 与 Morris-Lecar。原笔记将它们放进以下链条：linear two-neuron oscillator、complex eigenvalues、Euler formula、damped oscillations。

Why it follows. 完整求 two-neuron inhibitory/excitatory weight matrix 的 complex eigenvalues, 并用 Euler formula 解释 damped oscillatory components。其逻辑后果是：明确线性系统中的 complex eigenvalues 不自动产生 sustained limit cycle；衰减/增长由 real part 决定。后面的正式推导会逐项写出 assumptions、shape/units、limiting cases 与 failure conditions。

Figure reading. 把图与矩阵 shape 对齐：轴代表变量或 mode，椭圆方向对应 eigenvectors，轴长/速率对应 eigenvalues；不要把统计轴自动解释为因果通路。

Stop & Predict

half-center 的核心 connectivity?

4.3 Lecture20-CentralPatternGenerators.pdf • p. 3

Nonlinear oscillations:

Pacemaker neurons:

The Morris-Lecar model is a simple Hodgkin-Huxley-like model that is capable of spontaneous oscillations.

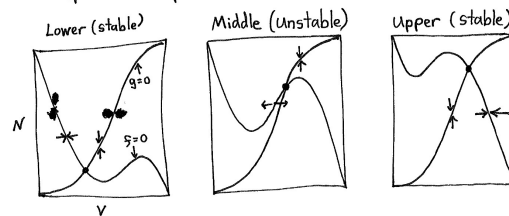
$$C \frac{dV}{dt} = -g_L(V - E_L) - \underbrace{\bar{g}_{Ca} M_\infty(V)(V - E_{Ca})}_{\text{Depolarizing calcium current}} - \underbrace{\bar{g}_K N(V - E_K)}_{\text{Hyperpolarizing potassium current}} + \underbrace{I_e}_{\text{External input}}$$

$$\tau_N \frac{dN}{dt} = -N + N_\infty(V)$$

Both M_∞ and N_∞ are sigmoidal curves. We will analyze the dynamics geometrically. First, we have

$$\frac{dV}{dt} = f(V, N), \quad \frac{dN}{dt} = g(V, N)$$

in abstract form, with V -nullcline $f(V, N) = 0$ and N -nullcline $g(V, N) = 0$. We plot them and find three fixed point types are possible because of "cubic-like" V -nullcline



Original-note anchor. [Source: Lecture20-CentralPatternGenerators.pdf, p. 3]

Clean reconstruction. Page 3: Morris-Lecar equations, V/N nullclines, fixed points.

What the note is saying. 本页的中心对象是 nullclines/limit cycle 与 release vs escape。原笔记将它们放进以下链条: Morris-Lecar equations, V/N nullclines, fixed points.

Why it follows. 逐项解释 Morris-Lecar voltage/recovery equations, sigmoidal steady-state functions, fast-slow separation。其逻辑后果是: 从 nullclines 和 fixed-point stability 解释 relaxation oscillation 的四阶段 trajectory。后面的正式推导会逐项写出 assumptions、shape/units、limiting cases 与 failure conditions。

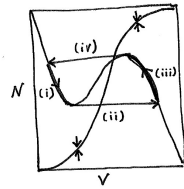
Figure reading. 读图时先标出两条 nullcline/坐标轴, 再判断向量场方向、交点稳定性与轨迹穿越顺序; 轨迹本身不等于因果机制。

Stop & Predict

complex eigenvalue 的 real part 决定什么?

4.4 Lecture20-CentralPatternGenerators.pdf • p. 4

The dynamics are oscillatory when the fixed point is unstable. This is easiest seen by noting that the V -dynamics are faster than the N -dynamics. The system thus relaxes quickly to the V -nullcline and follows it in the direction of dN/dt .



- (i) V increases as N decreases
- (ii) V jumps from end of lower branch to upper branch. (It can't go further as N decreases)
- (iii) V decreases as N increases
- (iv) V jumps from end of upper branch to lower branch.

As this cycle repeats, V oscillates autonomously



Half-center oscillators:

Two inhibitorially coupled Morris-Lecar neurons can oscillate even if the single neurons do not.

$$\begin{aligned}
 \textcircled{1} \quad C \frac{dV_1}{dt} &= C S(V_1, N_1) * -\bar{g}_s \underbrace{S_{\infty}(V_2)}_{\text{sigmoid}} (V_1 - E_s) \\
 C \frac{dV_2}{dt} &= C S(V_2, N_2) - \bar{g}_s S_{\infty}(V_1) (V_2 - E_s) \\
 \frac{dN_i}{dt} &= g(V_i, N_i)
 \end{aligned}$$

The magnitude of the synaptic conductance is a sigmoidal function of the pre-synaptic voltage. For

Original-note anchor. [Source: Lecture20-CentralPatternGenerators.pdf, p. 4]

Clean reconstruction. Page 4: relaxation oscillation cycle, inhibitory coupled Morris-Lecar half-center.

What the note is saying. 本页的中心对象是 rate-model analogue 与 pacemaker vs follower。原笔记将它们放进以下链条: relaxation oscillation cycle、inhibitory coupled Morris-Lecar half-center。

Why it follows. 从 nullclines 和 fixed-point stability 解释 relaxation oscillation 的四阶段 trajectory。其逻辑后果是: 分析 mutually inhibitory half-center with synaptic threshold; 逐步区分 intrinsic release 和 intrinsic escape。后面的正式推导会逐项写出 assumptions、shape/units、limiting cases 与 failure conditions。

Figure reading. 先识别图中对象、箭头、参数和坐标系, 再问它表达的是定义、推导、模型预测还是经验观察; 示意图不提供未标注的定量精度。

Stop & Predict

imaginary part 决定什么?

4.5 Lecture20-CentralPatternGenerators.pdf • p. 5

simplicity, we model it with a sharp threshold

$$S_{\infty}(V) = \begin{cases} 0, & V < V_T \\ 1, & V > V_T \end{cases}$$

When the presynaptic neuron's voltage is below the synaptic threshold, V_T , the postsynaptic neuron evolves freely. When above, the postsynaptic neuron is inhibited

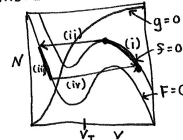
$$I_s = -\bar{g}_s (V - E_s)$$

synaptic current

$$\frac{dV}{dt} = S(V, N) - \frac{\bar{g}_s}{C} (V - E_s) = F(V, N)$$

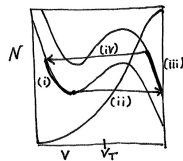
In a half-center oscillator, the two neurons alternate between being free and inhibited. The free neuron's nullclines are $S(V, N) = 0$, $g(V, N) = 0$. The inhibited neuron's nullclines are $F(V, N) = 0$, $g(V, N) = 0$.

Intrinsic release of inhibition mechanism



- (i) Free neuron decreases V as N increases
- (ii) Free neuron jumps to low voltage and releases inhibition at end of upper branch. Becomes inhibited.
- (iii) Inhibited neuron increases V as N decreases
- (iv) Inhibited neuron becomes free and jumps to F -nullcline.

Intrinsic escape from inhibition mechanism:



- (i) Inhibited increases V as N decreases
- (ii) Inhibited neuron jumps to high voltage and escapes inhibition at end of lower branch. Becomes Free.
- (iii) Free neuron decreases V as N increases
- (iv) Free neuron becomes inhibited and jumps to F -nullcline

Original-note anchor. [Source: Lecture20-CentralPatternGenerators.pdf, p. 5]

Clean reconstruction. Page 5: sharp synaptic threshold、intrinsic release 与 intrinsic escape.

What the note is saying. 本页的中心对象是 half-center 与 complex eigenvalues。原笔记将它们放进以下链条：sharp synaptic threshold、intrinsic release 与 intrinsic escape。

Why it follows. 分析 mutually inhibitory half-center with synaptic threshold；逐步区分 intrinsic release 和 intrinsic escape。其逻辑后果是：从 nullclines 和 fixed-point stability 解释 relaxation oscillation 的四阶段 trajectory。后面的正式推导会逐项写出 assumptions、shape/units、limiting cases 与 failure conditions。

Figure reading. 先识别图中对象、箭头、参数和坐标系，再问它表达的是定义、推导、模型预测还是经验观察；示意图不提供未标注的定量精度。

Stop & Predict

为什么笔记线性 network 不产生 sustained rhythm?

4.6 Lecture20-CentralPatternGenerators.pdf • p. 6

Half-center rate networks:

The same geometry of nullclines can produce half-center oscillators in firing rate models. Replace V with h and N with an adaptation/recovery variable.

Original-note anchor. [Source: Lecture20-CentralPatternGenerators.pdf, p. 6]

Clean reconstruction. Page 6: half-center firing-rate model extension.

What the note is saying. 本页的中心对象是 Morris-Lecar 与 nullclines/limit cycle。原笔记将它们放进以下链条：half-center firing-rate model extension。

Why it follows. 说明如何在 firing-rate model 中用 adaptation/recovery variable 复现同类 geometry。其逻辑后果是：分析 mutually inhibitory half-center with synaptic threshold；逐步区分 intrinsic release 和 intrinsic escape。后面的正式推导会逐项写出 assumptions、shape/units、limiting cases 与 failure conditions。

Figure reading. 先识别图中对象、箭头、参数和坐标系，再问它表达的是定义、推导、模型预测还是经验观察；示意图不提供未标注的定量精度。

Stop & Predict

Morris-Lecar 的两个 state?

5 补全推导与数学支架 / Derivations

5.1 linear two-neuron complex modes

【必要前置】 分析 $W = \begin{bmatrix} 0 & -w_0 \\ w_0 & 0 \end{bmatrix}$ 。

$$\lambda_{\pm} = \pm i|w_0|$$

$$\tau \dot{\alpha}_{\pm} = -(1 \mp i|w_0|)\alpha_{\pm}$$

$$\alpha_{\pm}(t) = \alpha_{\pm}(0)e^{-t/\tau}e^{\pm i|w_0|t/\tau}$$

1. characteristic polynomial $\lambda^2 + w_0^2 = 0$ 。
2. network ODE 还有 intrinsic $-I$ term, 所以 full mode exponent real part 为 $-1/\tau$ 。
3. Euler identity 把 complex exponential 分成 sine/cosine。
4. amplitude $e^{-t/\tau}$ 衰减, 因此不是 sustained oscillator。

Geometric/computational intuition. imaginary part 给 angular frequency, real part 给 envelope decay/growth。

Biological interpretation / failure condition. 看到 complex eigenvalues 就断言 limit cycle, 是线性/非线性混淆。

Micro-check

imaginary part 决定什么?

5.2 Morris-Lecar fast-slow equations

【补全推导】 用两个 variables 描述 intrinsic bursting/spiking geometry。

$$C\dot{V} = -g_L(V - E_L) - \bar{g}_{Ca}M_\infty(V)(V - E_{Ca}) - \bar{g}_K N(V - E_K) + I$$

$$\tau_N \dot{N} = N_\infty(V) - N$$

1. V 是 fast voltage, N 是 slower K activation/recovery。
2. V-nullcline 由 $dV/dt=0$, 常呈 cubic/N shape; N-nullcline 为 $N=N_\infty(V)$ 。
3. intersection 是 fixed point, 其 stability 决定 resting vs oscillatory regime。
4. fast-slow separation 让 trajectory 沿 stable branches 慢走, 在 fold 附近快跳。

Geometric/computational intuition. relaxation oscillator 在慢积累与快 transition 之间循环。
Biological interpretation / failure condition. 具体 nullcline shapes/parameters 不同会改变 bifurcation 与 oscillation type。

Micro-check

为什么笔记线性 network 不产生 sustained rhythm?

5.3 relaxation cycle 四阶段

【跨页连接】 按 phase-plane 解释 autonomous oscillation。

$$\dot{V} = F(V, N), \quad \dot{N} = G(V, N)$$

1. lower branch 上 N 下降或缓慢变化, V 低。
2. 达到 fold 后 fast V 跳到 upper branch。
3. upper branch 上 N 增加, K current 增强。
4. 达到另一 fold 后 V 快速回到 lower branch, cycle 重复。

Geometric/computational intuition. 一条变量慢慢改变系统的 instantaneous equilibrium, 最终触发快速 jump。
Biological interpretation / failure condition. phase-plane arrows 必须由 F/G signs 判断, 不能仅按图形顺时针猜。

Micro-check

Morris-Lecar 的两个 state?

5.4 half-center release 与 escape

【模型边界】 互抑 neurons 如何轮流 active?

$$I_{syn} = -\bar{g}_s S_{\infty}(V_{pre})(V - E_s)$$

$$S_{\infty}(V) \approx H(V - V_T)$$

1. active neuron 超过 V_T , 强抑制另一 neuron。
2. release: active neuron 的 intrinsic adaptation 使其离开 upper branch/停止, 解除对 partner 抑制。
3. escape: inhibited neuron 的 intrinsic recovery 使其越过 lower-branch threshold, 即使 inhibition 仍存在也跳出。
4. 两者可通过相位图中由哪一个 cell 的 intrinsic transition 先发生来区分。

Geometric/computational intuition. 交替可由“发放者放手”或“被抑制者逃脱”启动。

Biological interpretation / failure condition. 实际 circuits 可能混合 release/escape、sensory feedback 与 neuromodulation。

Micro-check

nullcline intersection 是什么?

6 Cross-page synthesis / 跨页因果与数学链

阻断 synapses 后 rhythm 是否保留，可区分 intrinsic pacemaker 与 network-generated rhythm 的候选机制。线性 two-neuron complex eigenvalues 只产生 damped 或 growing oscillatory components; 稳定的 autonomous limit cycle 需要 nonlinearity。Morris–Lecar 的 fast voltage/slow recovery nullclines 产生 relaxation oscillation; 互抑 half-center 可通过 intrinsic release 或 escape 交替。

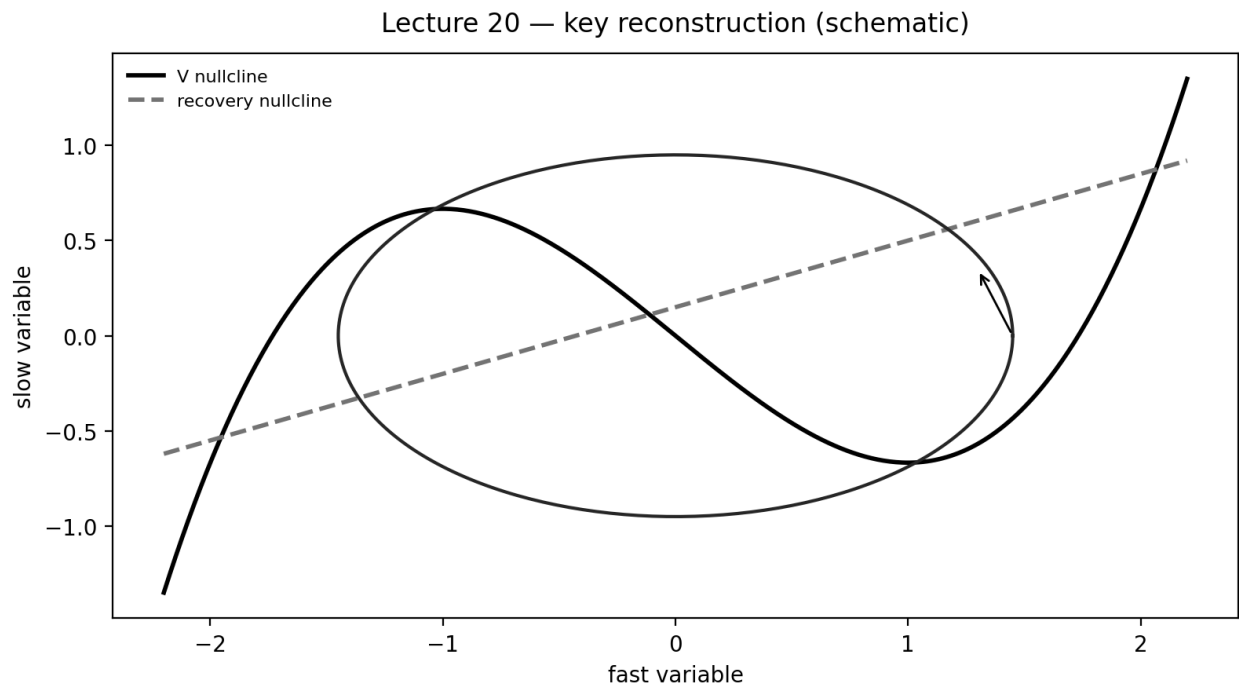


Figure note: schematic reconstruction; it encodes qualitative relations from the lesson and does not invent precise empirical data points.

7 Worked examples and transfer

7.1 synaptic block diagnostic

【原课/核心例题】 Prompt. 阻断 synapses 后 neuron 1 仍 rhythmic, neuron 2 变 tonic。

Worked solution. 支持 neuron 1 intrinsic pacemaker、neuron 2 follower; 但药物也可能改变 intrinsic currents, 仍需控制。

Check. network-generated half-center 通常在互抑被破坏后失去交替 rhythm。

7.2 complex eigenvalue reading

【跨课连接】 Prompt. full system eigenvalue 为 $-2 \pm 5i \text{ s}^{-1}$ 。

Worked solution. envelope time constant $1/2$ s, angular frequency 5 rad/s (约 0.80 Hz) ; 是 damped oscillation。

Check. real part 决定 stability, imaginary part 决定 rotation frequency。

Micro-check

Explain in your own words. 用不超过 120 字解释：本课从 source page 1 到最后一页，究竟把哪一个输入、状态、统计量或学习规则变成了可检验 prediction？

Self-reflection. 你最可能犯的是 concept、symbol、algebra、assumption 还是 causality error？写出一个具体例子。

8 Formula and notation sheet

1. complex mode

$$\lambda = a \pm ib$$

Conditions / conventions: a controls envelope, b frequency

2. Euler decomposition

$$e^{(a+ib)t} = e^{at}(\cos bt + i \sin bt)$$

Conditions / conventions: complex representation

3. two-dimensional conductance model

$$\dot{V} = F(V, N), \quad \dot{N} = G(V, N)$$

Conditions / conventions: fast-slow variables

4. V-nullcline

$$F(V, N) = 0$$

Conditions / conventions: $dV/dt=0$

5. N-nullcline

$$G(V, N) = 0$$

Conditions / conventions: $dN/dt=0$

6. sharp synaptic threshold approximation

$$S_{\infty}(V) = H(V - V_T)$$

Conditions / conventions: idealized switch

9 Bilingual glossary

中文	English / symbol	Operational meaning
中央模式发生器	central pattern generator, CPG	可在无周期 sensory drive 时产生 rhythmic motor pattern 的 neural circuit。
起搏细胞	pacemaker neuron	依 intrinsic dynamics 自发 rhythmic。
半中心振荡器	half-center oscillator	两个互抑 units 交替 active 的 network oscillator。
复特征值	complex eigenvalue	同时编码 exponential envelope 与 oscillatory phase 的 eigenvalue。
Morris-Lecar 模型	Morris-Lecar model	简化 conductance-based two-variable neuron model。
相平面	phase plane	二维 state space 中的 trajectories/nullclines。
松弛振荡	relaxation oscillation	slow drift 与 fast jumps 交替的 limit cycle。
内在释放	intrinsic release	active neuron 停止后解除 inhibition。
内在逃逸	intrinsic escape	inhibited neuron 先恢复并跳出 inhibition。
极限环	limit cycle	吸引附近 trajectories 的 closed periodic orbit。

10 Common traps, assumptions, and limitations

1. 所有 rhythmic activity 都称 CPG。
2. complex eigenvalues 自动等于 sustained oscillation。
3. 忽略 full eigenvalue 的 real part。
4. nullcline 当作 trajectory。
5. fixed point intersection 存在就认为 stable。
6. release 与 escape 只按表面 firing order 区分，而不看哪个 intrinsic transition 触发 switch。
7. 把 sensory feedback 排除在真实 locomotor rhythm 外。

Course-specific risk boundary. 不要把所有 rhythmic activity 都归为 CPG，也不要把 damped linear oscillation 与 autonomous nonlinear oscillator 混为一谈。

11 Cumulative Knowledge Check

Do not turn to the Hint Bank until you have written a first attempt.

1. synapse block 后 rhythm 保留支持什么?

2. half-center 的核心 connectivity?

3. complex eigenvalue 的 real part 决定什么?

4. imaginary part 决定什么?

5. 为什么笔记线性 network 不产生 sustained rhythm?

6. Morris-Lecar 的两个 state?

7. nullcline intersection 是什么?

8. relaxation oscillation 为什么有 fast jump?

9. release 由谁先变化?

10. escape 由谁先变化?

11. sharp threshold S_∞ 的作用?

12. damped oscillation 与 limit cycle 的区别?

13. 公式表中的“complex mode” 解决什么问题? 写出适用条件与一个 sanity check。

14. 公式表中的“Euler decomposition” 解决什么问题? 写出适用条件与一个 sanity check。

15. 公式表中的“two-dimensional conductance model” 解决什么问题? 写出适用条件与一个 sanity check。

16. 公式表中的“V-nullcline” 解决什么问题？ 写出适用条件与一个 sanity check。

17. 公式表中的“N-nullcline” 解决什么问题？ 写出适用条件与一个 sanity check。

12 Hint Bank

1. **Hint 1.** 先识别对象、变量与假设。
Hint 2. 写出最相关的定义或方程，再代入。
Hint 3. 检查单位、shape、符号与极限情况。
2. **Hint 1.** 先识别对象、变量与假设。
Hint 2. 写出最相关的定义或方程，再代入。
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12. **Hint 1.** 先识别对象、变量与假设。
Hint 2. 写出最相关的定义或方程，再代入。
Hint 3. 检查单位、shape、符号与极限情况。
13. **Hint 1.** 先从“complex mode”对应的变量关系入手。
Hint 2. 把条件“a controls envelope, b frequency”逐字转换为 assumption。
Hint 3. 写公式，再检查至少一种极端参数或单位。
14. **Hint 1.** 先从“Euler decomposition”对应的变量关系入手。
Hint 2. 把条件“complex representation”逐字转换为 assumption。
Hint 3. 写公式，再检查至少一种极端参数或单位。

15. **Hint 1.** 先从 “two-dimensional conductance model” 对应的变量关系入手。
Hint 2. 把条件 “fast-slow variables” 逐字转换为 assumption。
Hint 3. 写公式, 再检查至少一种极端参数或单位。
16. **Hint 1.** 先从 “V-nullcline” 对应的变量关系入手。
Hint 2. 把条件 “ $dV/dt=0$ ” 逐字转换为 assumption。
Hint 3. 写公式, 再检查至少一种极端参数或单位。
17. **Hint 1.** 先从 “N-nullcline” 对应的变量关系入手。
Hint 2. 把条件 “ $dN/dt=0$ ” 逐字转换为 assumption。
Hint 3. 写公式, 再检查至少一种极端参数或单位。

13 Complete Answer Key with reasoning

1. 支持 cell-intrinsic pacemaker, 但仍需控制药物副作用。
2. mutual inhibition。
3. amplitude decay/growth 与 stability。
4. oscillation angular frequency。
5. full modes real part 为 $-1/\tau$, amplitude 衰减。
6. voltage V 与 recovery/ K activation N 。
7. fixed point。
8. slow branch 到 fold 后局部 stable equilibrium 消失, fast V dynamics 跳到另一 branch。
9. active neuron intrinsic adaptation 使其停止。
10. inhibited neuron intrinsic recovery 使其跳出。
11. 把 presynaptic voltage 对 synaptic activation 简化成开/关。
12. 前者 amplitude 回到 fixed point; 后者收敛到非零 closed orbit。
13. 适用条件/约定: a controls envelope, b frequency. sanity check 应检查单位、符号、极限或 shape 是否与定义一致。

$$\lambda = a \pm ib$$

14. 适用条件/约定: complex representation. sanity check 应检查单位、符号、极限或 shape 是否与定义一致。

$$e^{(a+ib)t} = e^{at}(\cos bt + i \sin bt)$$

15. 适用条件/约定: fast-slow variables. sanity check 应检查单位、符号、极限或 shape 是否与定义一致。

$$\dot{V} = F(V, N), \dot{N} = G(V, N)$$

16. 适用条件/约定: $dV/dt=0$. sanity check 应检查单位、符号、极限或 shape 是否与定义一致。

$$F(V, N) = 0$$

17. 适用条件/约定: $dN/dt=0$. sanity check 应检查单位、符号、极限或 shape 是否与定义一致。

$$G(V, N) = 0$$

14 Source-page concordance and coverage audit

Source file	Page	Coverage status
Lecture20-CentralPatternGenerators.pdf	1	Main source-page unit: thumbnail, reconstruction, explanation, prediction question.
Lecture20-CentralPatternGenerators.pdf	2	Main source-page unit: thumbnail, reconstruction, explanation, prediction question.
Lecture20-CentralPatternGenerators.pdf	3	Main source-page unit: thumbnail, reconstruction, explanation, prediction question.
Lecture20-CentralPatternGenerators.pdf	4	Main source-page unit: thumbnail, reconstruction, explanation, prediction question.
Lecture20-CentralPatternGenerators.pdf	5	Main source-page unit: thumbnail, reconstruction, explanation, prediction question.
Lecture20-CentralPatternGenerators.pdf	6	Main source-page unit: thumbnail, reconstruction, explanation, prediction question.

15 Errata, uncertainty log, and external endnotes

15.1 Errata / source cautions

1. Page 2 的线性 oscillations 含 $e^{-t/\tau}$ envelope, 是 damped components, 不是 autonomous sustained CPG。
2. Page 3–4 的 Morris–Lecar nullclines 为 schematic; 具体 shape/stability 依参数。
3. Page 6 仅指出 firing-rate model 可复现相同 geometry, 未给唯一方程。

15.2 Uncertainty log

No unresolved handwriting uncertainty changes a core definition, equation, figure interpretation, or answer in this companion. Where handwriting was potentially ambiguous, the source-page image is preserved and the reconstruction avoids claiming unverified detail.

15.3 External endnotes

No external web or textbook source was used to replace the uploaded notes. Added material is limited to necessary mathematical scaffolding, explicit assumptions, standard sanity checks, and clearly labeled transfer examples.

15.4 Final self-audit

All source pages listed above were visually inspected. Equations were checked for symbol definitions, units/shapes, assumptions, algebraic transitions, limiting cases, and failure conditions. The PDF is static and contains no interactive form fields or scripts.